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Walker

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(54) **INTEGRATED CIRCUIT DEVICE
INCLUDING AN SRAM PORTION HAVING
END POWER SELECT CIRCUITS**

(71) Applicant: **SAMSUNG ELECTRONICS CO.,
LTD.**, Suwon-si (KR)

(72) Inventor: **Darryl G. Walker**, San Jose, CA (US)

(73) Assignee: **SAMSUNG ELECTRONICS CO.,
LTD.**, Suwon-si (KR)

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G11C 11/412 (2006.01)
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(2013.01); **G11C 11/418** (2013.01)

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G11C 7/22; G11C 11/4074; G11C 11/417;
G11C 2207/2227; G11C 5/02; G11C
5/025; G11C 5/148; G11C 7/00; G11C
7/10; G11C 8/00; G11C 11/40615;
(Continued)

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application claims priority.

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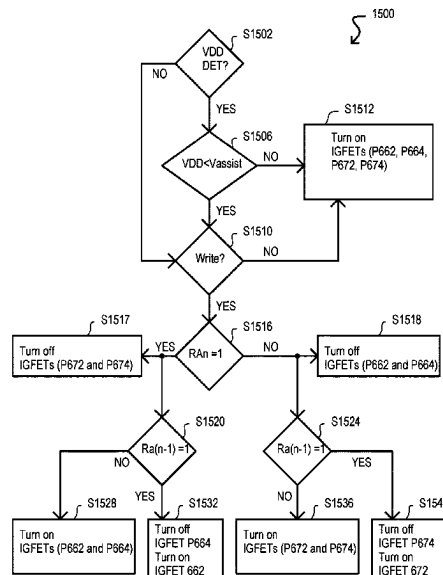
Primary Examiner — Thong Q Le

(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57) **ABSTRACT**

An integrated circuit device that has improved write margin
at low operating voltages is disclosed. The integrated circuit
device can include an SRAM array that has end power select
circuits that can include selection circuits that provide a
controllable impedance path between a power supply poten-
tial and an array power line. A power supply detection circuit
may provide an assist enable signal when a power supply
potential is low enough that write assist is needed. A power
control circuit may provide end power control signals to end
power select circuits to selectively control an impedance
path between a power supply potential and an array power
line to provide an I-R drop to a selected memory cell. In this
way, write margins may be improved at low operating
voltages.

20 Claims, 10 Drawing Sheets



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(60) Provisional application No. 63/244,442, filed on Sep. 15, 2021.

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G11C 11/418 (2006.01)

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(58) **Field of Classification Search**

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13/0038; G11C 13/004; G11C 13/0069; G11C 2013/0045; G11C 2013/0078; G11C 2029/0407; G11C 2207/104; G11C 2213/15; G11C 29/021; G11C 29/028; G11C 29/12; G11C 29/12005; G11C 29/46; G11C 29/787; G11C 5/145

See application file for complete search history.

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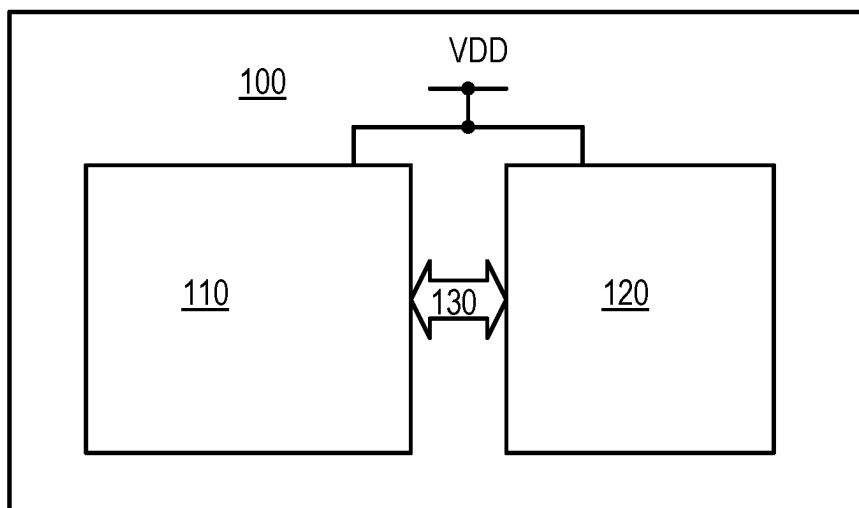


FIG. 1

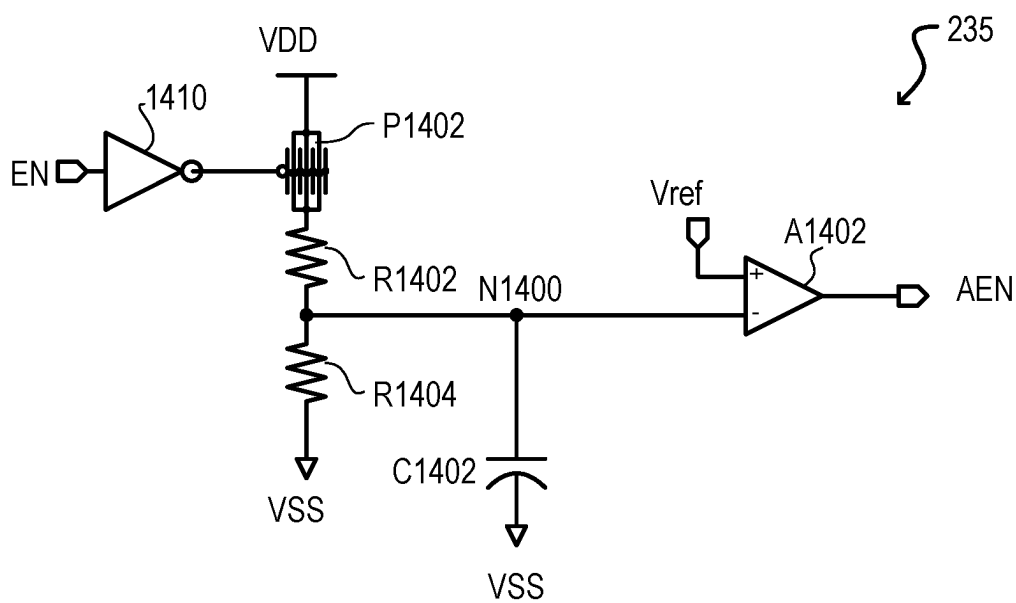


FIG. 14

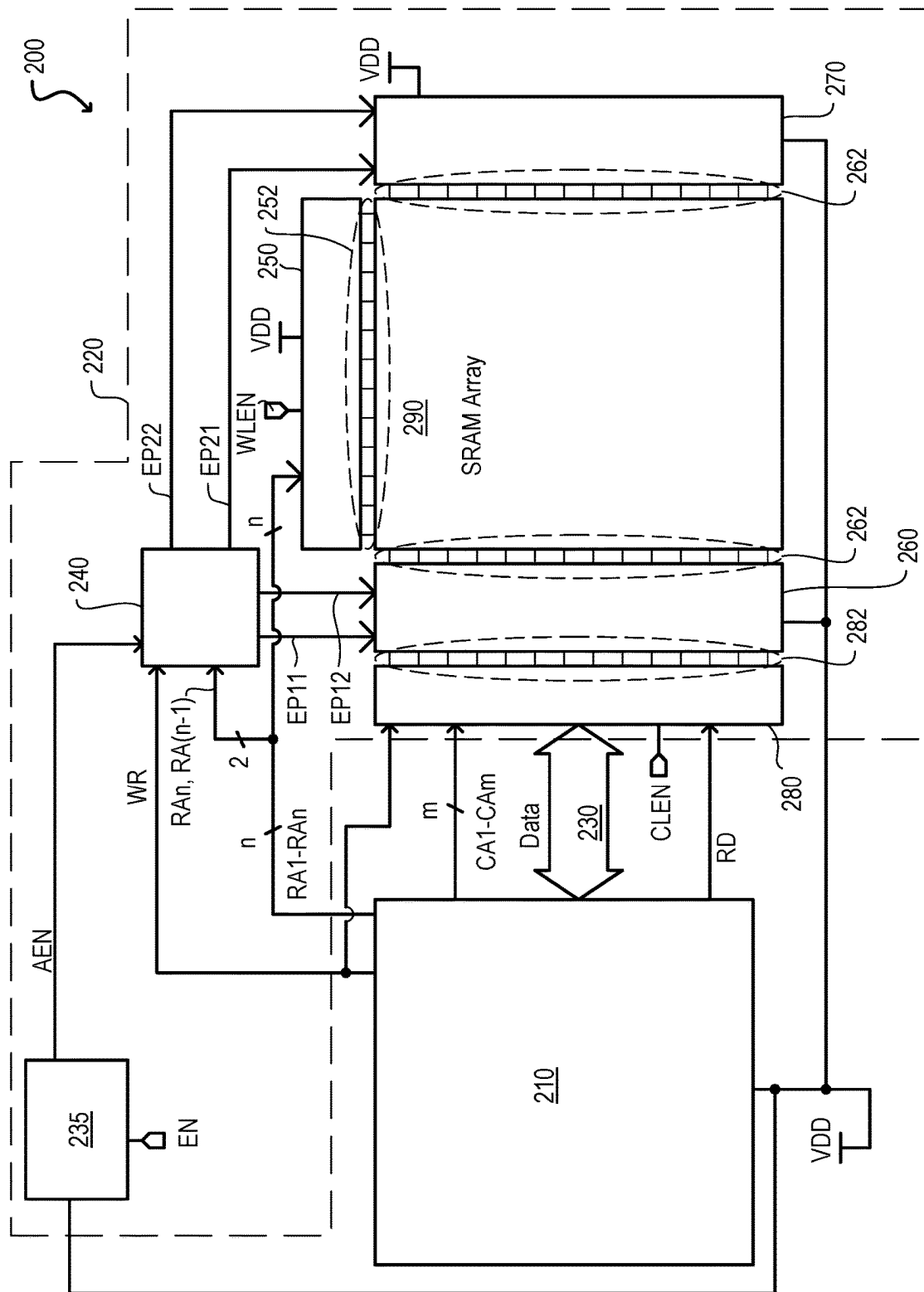


FIG. 2

FIG. 3A

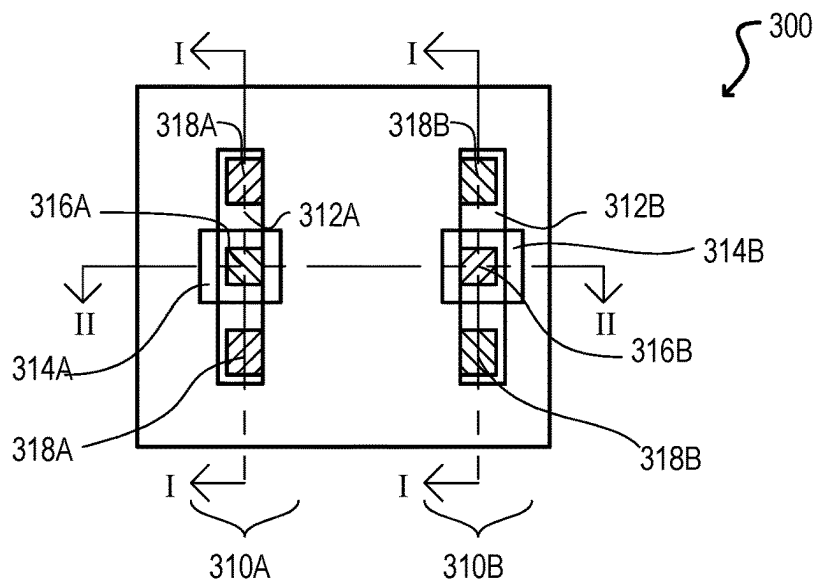


FIG. 3B

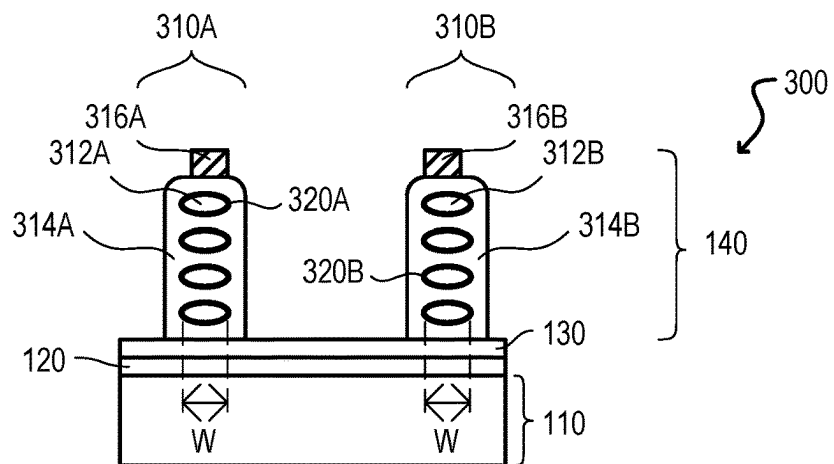
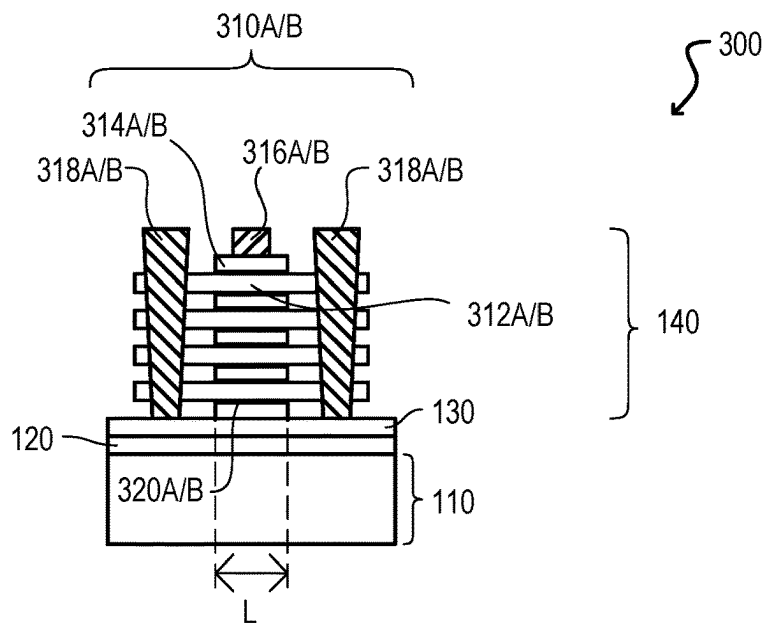


FIG. 3C



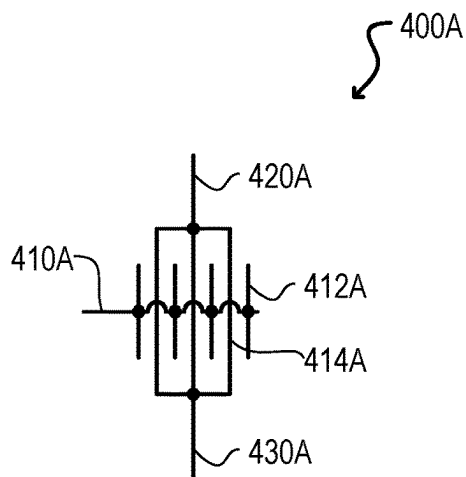


FIG. 4A

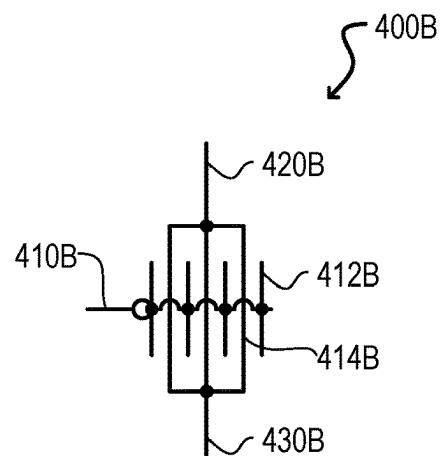


FIG. 4B

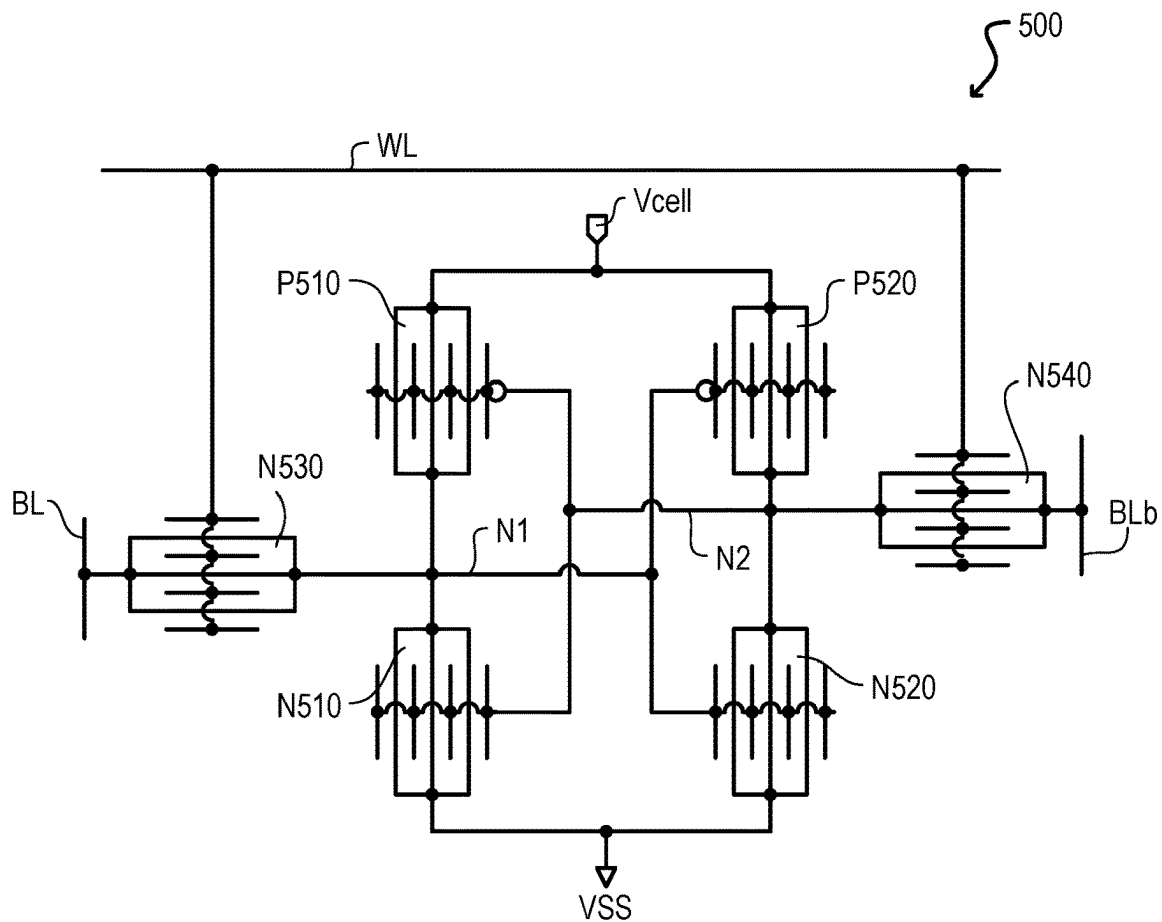


FIG. 5

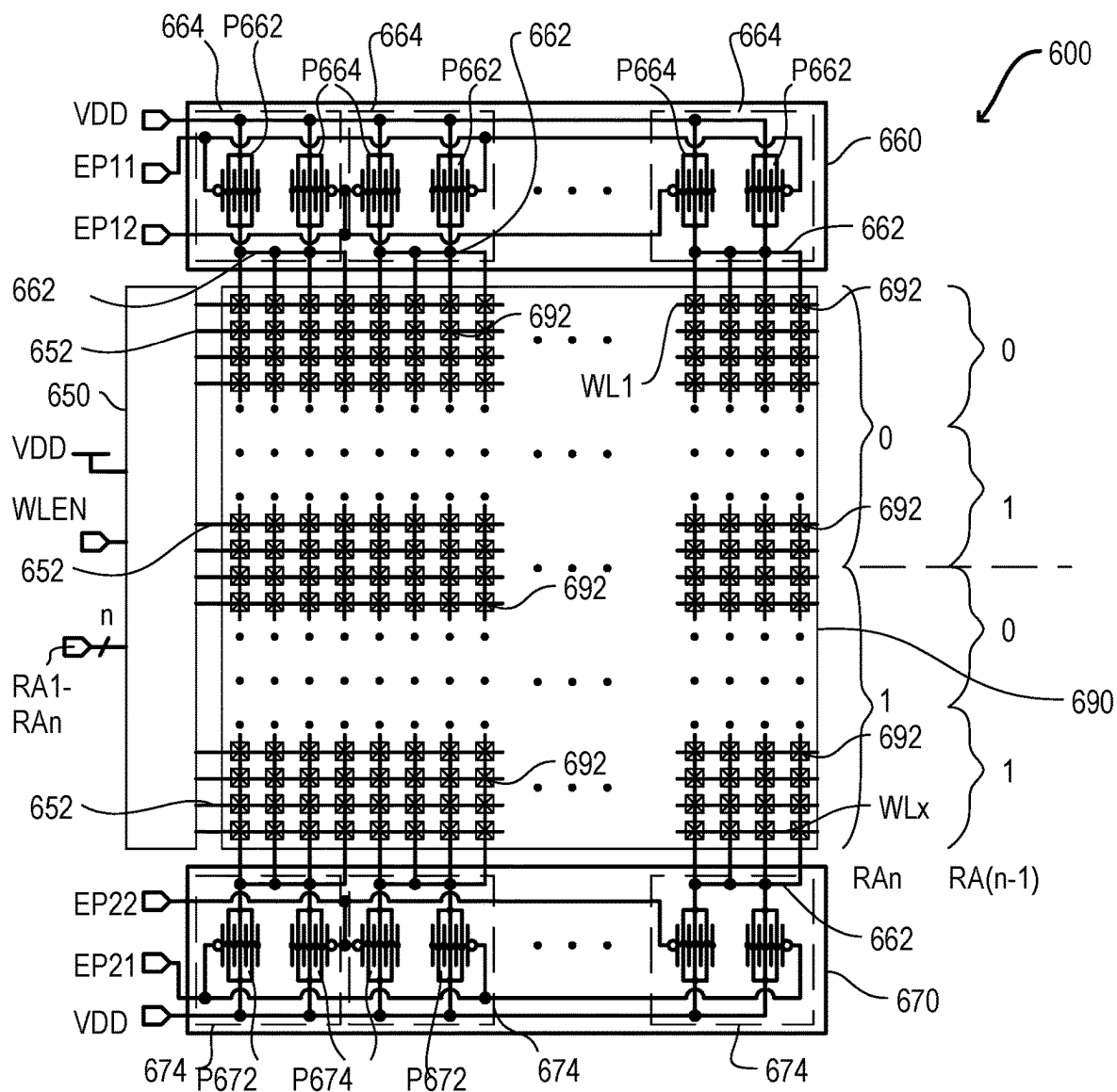


FIG. 6

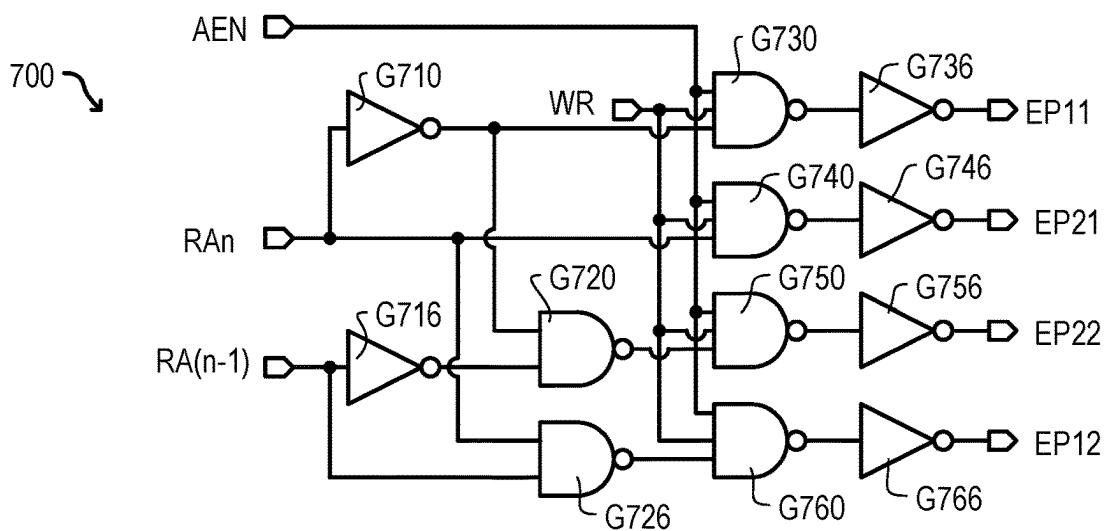


FIG. 7

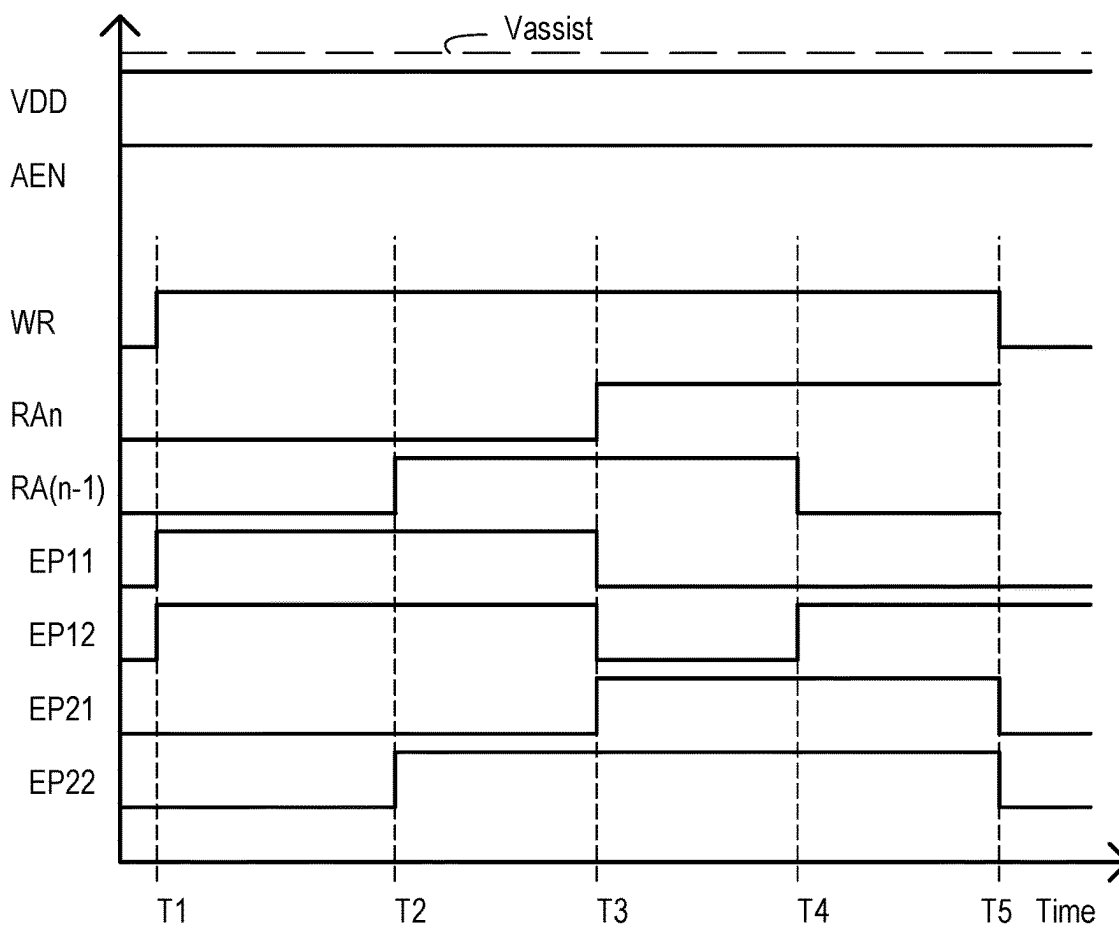


FIG. 8

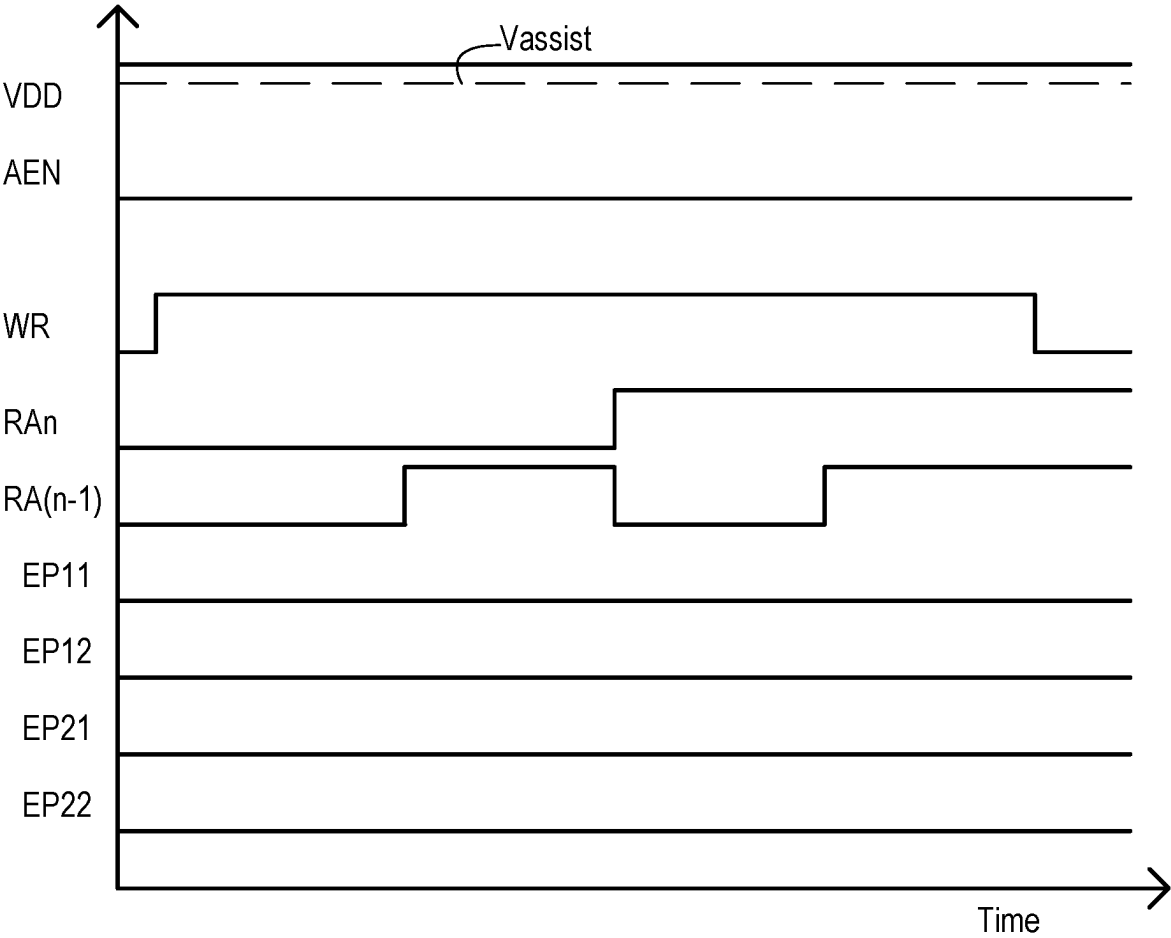


FIG. 9

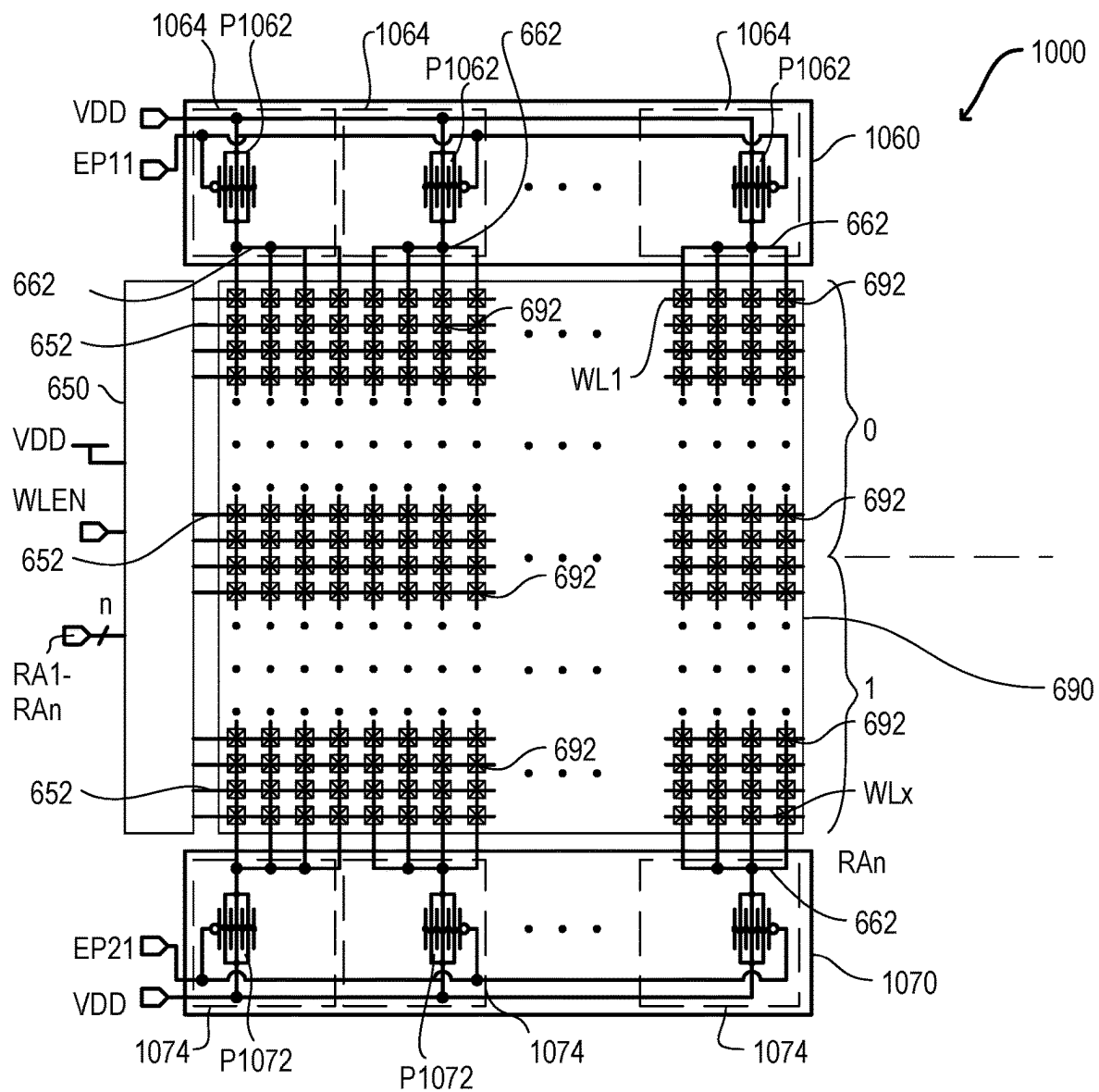


FIG. 10

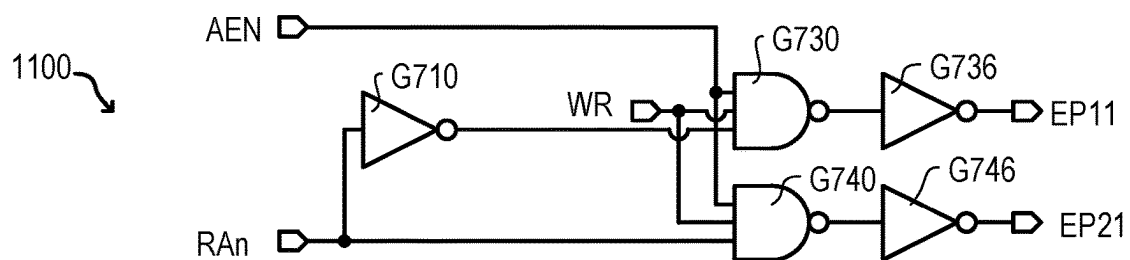


FIG. 11

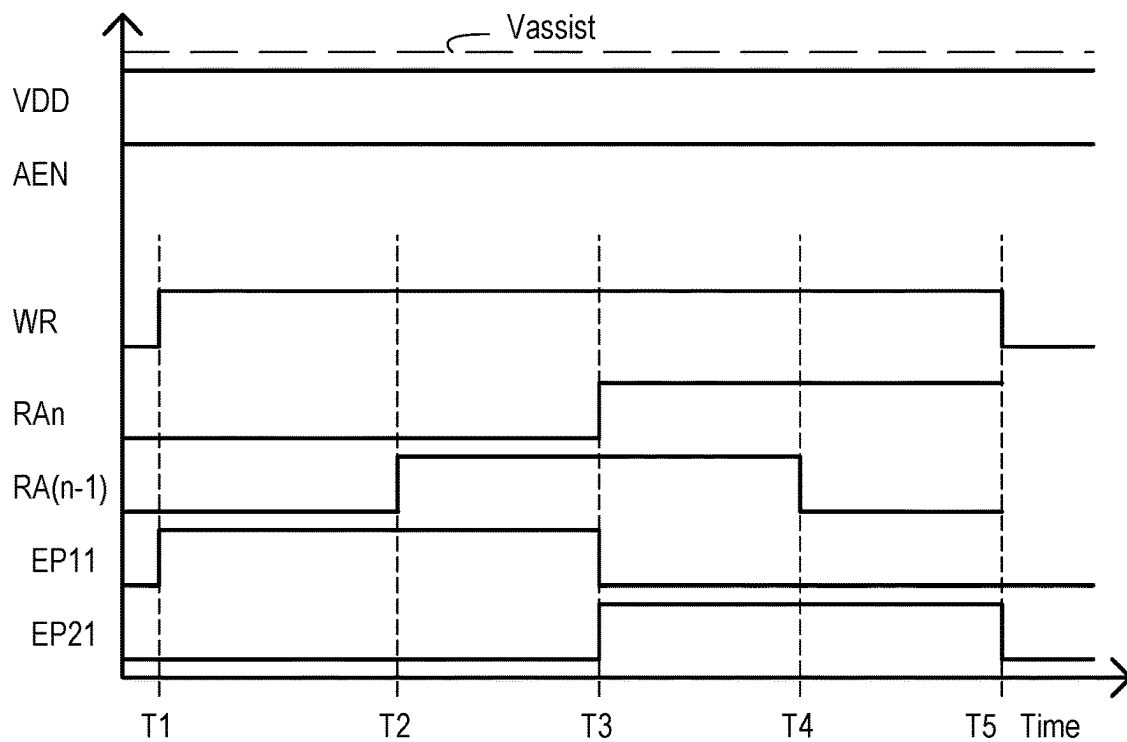


FIG. 12

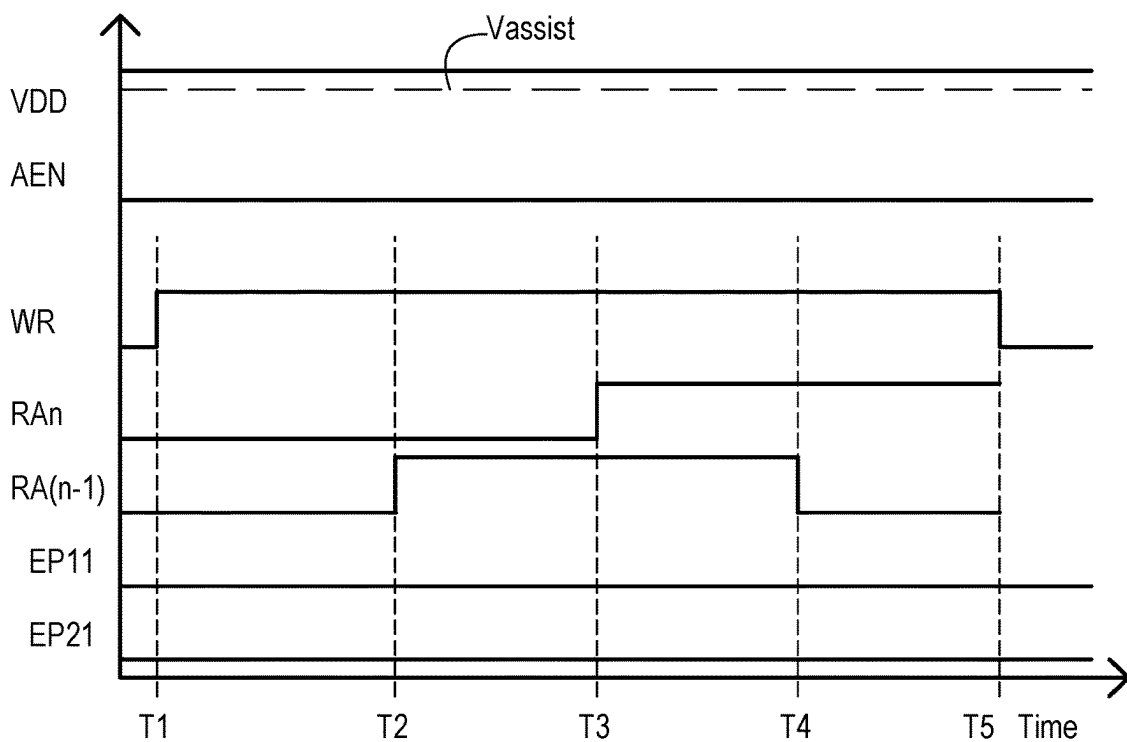


FIG. 13

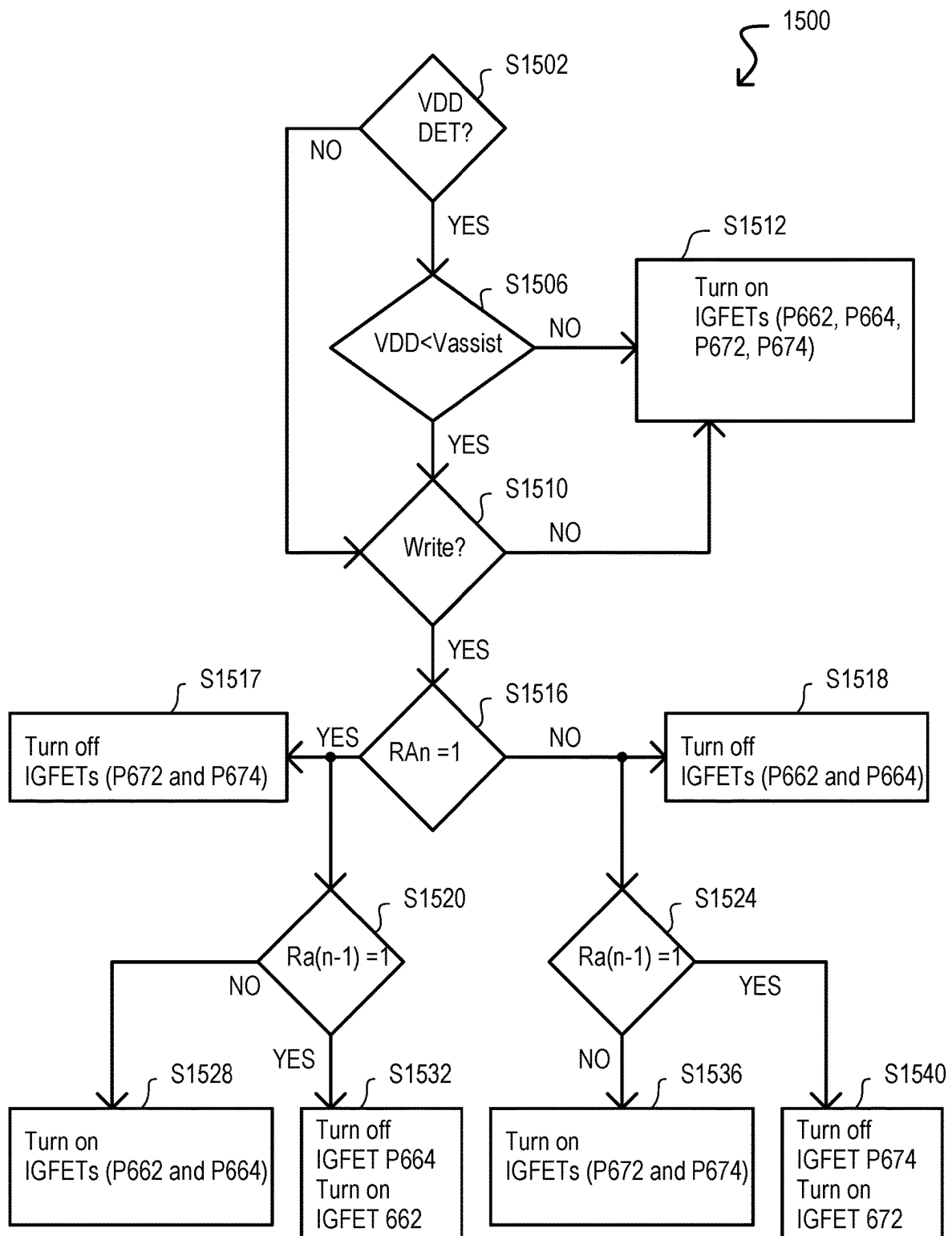


FIG. 15

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INTEGRATED CIRCUIT DEVICE INCLUDING AN SRAM PORTION HAVING END POWER SELECT CIRCUITS

This application is a continuation of U.S. patent application Ser. No. 17/738,321 filed May 6, 2022, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/244,442, filed Sep. 15, 2021, the contents all of which are incorporated by reference herein.

TECHNICAL FIELD

The present invention relates generally to a semiconductor device, and more particularly to improving efficient and reliable operation at low voltages of a static random access memory (SRAM) integrated circuit device.

BACKGROUND OF THE INVENTION

A static random access memory (SRAM) device may have difficulty reliably writing data when operating at various power supply potentials. In light of the above, it would be desirable to provide a method of improving the reliability of writing data to an SRAM memory cell over a wide range of operating conditions including power supply voltage levels.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block schematic diagram of an integrated circuit device according to an embodiment.

FIG. 2 is a block schematic diagram of an integrated circuit device according to an embodiment.

FIG. 3A is a top plan view of an integrated circuit device including transistors according to an embodiment.

FIG. 3B is a cross sectional view of an integrated circuit device including transistors according to an embodiment.

FIG. 3C is a cross sectional view of an integrated circuit device including transistors according to an embodiment.

FIGS. 4A and 4B are circuit schematic diagrams of complementary insulated gate field effect transistors having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment.

FIG. 5 is a circuit schematic diagram of an SRAM memory cell according to an embodiment.

FIG. 6 is a circuit schematic diagram of an integrated circuit device portion including an SRAM portion according to an embodiment.

FIG. 7 is a circuit schematic diagram of an array power control circuit according to an embodiment.

FIG. 8 is a logic timing diagram of an operation of the integrated circuit device when the power supply potential is below an assist potential according to an embodiment.

FIG. 9 is a logic timing diagram of an operation of the integrated circuit device when the power supply potential is above an assist potential according to an embodiment.

FIG. 10 is an integrated circuit device portion including an SRAM portion having end power supply circuits providing a write assist according to an embodiment.

FIG. 11 is a circuit schematic diagram an array power control circuit according to an embodiment.

FIG. 12 is a logic timing diagram of an operation of the integrated circuit device when the power supply potential is below an assist potential according to an embodiment.

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FIG. 13 is a logic timing diagram of an operation of the integrated circuit device when the power supply potential is above an assist potential according to an embodiment.

FIG. 14 is a circuit schematic diagram of a power supply level detection circuit according to an embodiment.

FIG. 15 is a flow chart diagram illustrating a method of operating an integrated circuit device according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

According to the embodiments set forth below, an integrated circuit device can include an SRAM array that has end power select circuits that can include selection circuits that provide a controllable impedance path between a power supply potential and an array power line. A power supply detection circuit may provide an assist enable signal when a power supply potential is low enough that write assist is needed. A power control circuit may provide end power control signals to end power select circuits to selectively control an impedance path between a power supply potential and an array power line to provide an I-R drop to a selected memory cell. In this way, write margins may be improved at low operating voltages. The integrated circuit device may include a processor circuit with on die SRAM.

Referring now to FIG. 1, an integrated circuit device according to an embodiment is set forth in a block schematic diagram and given the general reference character 100.

Integrated circuit device 100 may include a processor circuit portion 110 and a static random access memory (SRAM) circuit portion 120. Processor circuit portion 110 and SRAM circuit portion 120 may be powered by power supply potential VDD. Power supply potential VDD may be provided from external to integrated circuit device 100.

Processor circuit portion 110 may transmit and receive data to/from SRAM circuit portion 120 along a bus 130. Bus 130 may also be used to transmit command signals from processor circuit portion 110 to SRAM circuit portion 120. Examples of commands may be a read command and/or a write command. A read command may be used to receive data from SRAM circuit portion 120 that is stored in one or more SRAM memory cells and a write command may be used to provide data to SRAM circuit 120 to be stored in one or more SRAM memory cells. Processor circuit portion 110 may use data from SRAM circuit portion 120 for operational or computational purposes.

It may be desirable to operate processor circuit portion 110 and SRAM circuit portion 120 using the same power supply potential VDD. In this way, the same process node may be used and design complications of providing more than one power supply potential can be avoided, which also may decrease the package footprint of semiconductor device and system level design complications.

According to an embodiment, SRAM circuit portion 120 may have assist circuitry and techniques to improve data write margins. In this way power supply potential VDD may be shared with both processor circuit portion 110 and SRAM circuit portion 120.

SRAM circuit portion 120 and processor circuit portion 110 may be formed on one contiguous device, such as an integrated circuit formed on a chip on a contiguous wafer to be separated by a dicing technique to form a device, such as a semiconductor device forming integrated circuit device 100.

Referring now to FIG. 2, a more detailed schematic diagram of an integrated circuit device according to an

embodiment is set forth in a block schematic diagram and given the general reference character **200**.

Integrated circuit device **200** can include similar constituents as integrated circuit device **100** and such constituents may have the same reference character except beginning with a "2" instead of a "1".

Integrated circuit device **200** of FIG. 2 includes processor circuit portion **210** and SRAM circuit portion **220**. Processor circuit portion **210** and SRAM circuit portion **220** may commonly receive power supply potential VDD. Power supply potential VDD may be provided external from integrated circuit device **200**.

Processor circuit portion **210** may provide control and address signals to SRAM circuit portion. Such control and address signals may include a write signal WR, read signal RD, row address signals (RA1-RA_n), and column address signals (CA1 to CA_m), where n and m are integers. SRAM circuit portion **220** may operate as to receive or provide data on a data bus **230** from or to processor circuit portion **210**. In this way, processor instructions and/or data may be received and provided between processor circuit portion **210** and SRAM circuit portion **220**.

SRAM circuit portion **220** may include an SRAM array **290**, a power supply level detection circuit **235**, an array power control circuit **240**, a row decoder circuit **250**, end power select circuits (**260** and **270**), and a data interface circuit **280**.

Power supply level detection circuit **235** may receive an enable signal EN and power supply potential VDD and may provide an assist enable signal AEN. When enable signal EN is at an enable logic level (logic high), power supply detection circuit **235** may provide assist enable signal AEN having a logic high (assist enable logic level) in response to the power supply potential VDD being below an assist potential V_{assist} and power supply detection circuit **235** may provide assist enable signal AEN having a logic low (assist disable logic level) in response to the power supply potential VDD being above an assist potential V_{assist}. When enable signal is at a disable logic level (logic low), power supply level detection may provide assist enable signal AEN at a logic high (assist enable level).

Array power control circuit **240** may receive assist enable signal AEN, write signal WR, and row address bits (RA_n, RA(n-1)). Row address bits (RA_n, RA(n-1)) may be the two most significant row address bits received by row decoder circuit **250**. Array power control circuit **240** may provide end power control signals (EP11, EP12, EP21, and EP22) as outputs.

Row decoder circuit **250** may receive row address signals RA1-RA_n, a word line enable signal WLEN, and power supply potential VDD and may provide a word line signal on each word line of a plurality of word lines **252**. Only one of the plurality of word lines **252** may be active at a given time.

End power select circuit **270** may receive power supply potential VDD and end power select signals (EP21 and EP22). End power select circuit **260** may receive power supply potential VDD and end power select signals (EP21 and EP22). Data interface circuit **280** may receive column address signals (CA1-CA_m) to operate a column selection circuit portion of end data interface circuit **280**. Data interface circuit **280** may receive write signal WR and a read signal RD to operate a data read/write portion of data interface circuit **280**. Data interface circuit **280** may be electrically connected to SRAM array **290** through a plurality of bit lines **282**.

End power select circuits (**260** and **270**) may provide write assistance based on a location of a selected word line

of the plurality of word lines **252** by providing a power supply potential VDD from a predetermined end of the SRAM array **290** and by selecting a particular impedance path to modify an impedance between the power supply potential VDD and the SRAM array **290**.

End power select circuit (**260** and **270**) may be electrically connected to SRAM array **290** along array power supply lines **262**. Array power supply lines **262** may be parallel to bit lines **282** across the SRAM array **290**.

Processor circuit portions (**110** and **210**) and SRAM circuit portions (**120** and **220**) of integrated circuit device (**100** and **200**), respectively, may include a plurality of insulated gate field effect transistors (IGFETs). Each IGFET may include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate.

A description of the plurality of transistors on integrated circuit devices (**100** and **200**) will now be discussed with reference to FIGS. 3A to 3C.

Referring now to FIG. 3A, a top plan view of an integrated circuit device including transistors according to an embodiment is set forth and given the general reference character **300**.

Integrated circuit device **300** may include an N-type insulated gate field effect transistor (IGFET) **310A** and a P-type IGFET **310B**.

N-type IGFET **310A** and P-type IGFET **310B** may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate.

N-type IGFET **310A** may include drain/source contacts **318A**, a gate contact **316A**, a gate structure **314A**, and vertically aligned and horizontally disposed channel region structures **312A**.

P-type IGFET **310B** may include drain/source contacts **318B**, a gate contact **316B**, a gate structure **314B**, and vertically aligned and horizontally disposed channel region structures **312B**.

Referring now to FIG. 3B, a cross sectional view of integrated circuit device **300** according to an embodiment is set forth. The cross-sectional view is along the line II-II of FIG. 3A.

Integrated circuit device **300** may include a N-type IGFET **310A**, and a P-type IGFET **310B**.

N-type IGFET **310A** may include a gate contact **316A**, a gate structure **314A**, and vertically aligned and horizontally disposed channel regions **312A**, and gate insulating layer **320A**. Gate insulating layer **320A** may surround each vertically aligned and horizontally disposed channel regions **312A**.

P-type IGFET **310B** may include a gate contact **316B**, a gate structure **314B**, and vertically aligned and horizontally disposed channel regions **312B**, and gate insulating layer **320B**. Gate insulating layer **320B** may surround each vertically aligned and horizontally disposed channel regions **312B**.

Gate structures (**314A** and **314B**) are each contiguous gate structures that surround channel regions (**312A** and **312B**), respectively.

Referring now to FIG. 3C, a cross sectional view of integrated device **300** is set forth. The cross-sectional view is along the line I-I of FIG. 3A. As shown in FIG. 3A, there are two lines I-I as the N-type IGFET **310A** and P-type IGFET **310B** may have similar structures except the materials and/or doping of materials may differ and elements are designated with the suffix "A/B" to illustrate such. Semiconductor device **300** may include N-type and P-type

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IGFETs (310A/B) formed in region 140 above regions (110, 120, and 130). IGFET 310A/B may include a gate contact 316A/B, a gate structure 314A/B, vertically aligned and horizontally disposed channel regions 312A/B, gate insulating layer 320A/B, and drain/source contacts 318A/B. Gate structure 316A/B and gate insulating layer 320A/B may surround each vertically aligned and horizontally disposed channel regions 312A/B.

Drain/source contacts (318A/B) are commonly shared by the plurality of channel regions 312A/B, respectively to form common drain/source terminals for each IGFET (310A and 310B).

IGFETs (310A and 310B) may be formed by forming a layered crystal of two materials over region 130. For example, layers of silicon and silicon germanium may be formed. An etch and deposit step may then be used to form the source/drain regions (318A and 318B) may be formed. The silicon layer may form the channel regions (312A and 312B). After a vertical etch, the silicon germanium layers may be etched by using a chemical that can selectively etch silicon germanium with the source/drain regions (318A and 318B) used as support structures. Next, the gate dielectric layers (320A and 320B) may be formed using atomic layer deposition, for example of hafnium-dioxide. Then gate structure (316A and 316B) may be formed using atomic layer deposition of a metal layer, for example, tungsten. The n-type IGFETs 310A may have source/drain regions 318A doped with n-type carriers, such as phosphorous and/or arsenic, for example. The p-type IGFETs 310B may have source/drain regions 318B doped with p-type carriers, such as boron, for example. IGFETs (310A and 310B) may have a gate length L (i.e. channel length) of less than about 10 nm and may preferably have a gate length L of less than about 6 nm. IGFETs (310A and 310B) may have a gate width W (i.e. channel width) of less than about 10 nm and may preferably have a gate width W of less than about 6 nm.

Referring now to FIGS. 4A and 4B, circuit schematic diagrams of complementary IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment are set forth. FIG. 4A is a N-channel IGFET 400A and FIG. 4B is a P-channel IGFET 400B.

N-channel IGFET 400A includes a control gate terminal 410A, a first source/drain terminal 420A, and a second source/drain terminal 430A. Control gate terminal 410A may be electrically connected to control gate 412A. Control gate 412A may be drawn as a plurality of control gates on each side of a plurality of channel region 414A. In reality, control gate 412A may surround a plurality of horizontally disposed channel regions 414A that can be vertically aligned above a substrate. Each channel region 414A may form a controllable impedance path between first source/drain terminal 420A, and second source/drain terminal 430A. Control gate 412A may provide control to the controllable impedance path based on a threshold voltage for distinguishing between a high impedance path and a low impedance path.

P-channel IGFET 400B includes a control gate terminal 410B, a first source/drain terminal 420B, and a second source/drain terminal 430B. Control gate terminal 410B may be electrically connected to control gate 412B. Control gate 412B may be drawn as a plurality of control gates on each side of a plurality of channel region 414B. In reality, control gate 412B may surround a plurality of horizontally disposed channel regions 414B that can be vertically aligned above a substrate. Each channel region 414B may form a

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controllable impedance path between first source/drain terminal 420B, and second source/drain terminal 430B. Control gate 412B may provide control to the controllable impedance path based on a threshold voltage for distinguishing between a high impedance path and a low impedance path.

It is understood throughout the FIGS. (figures), any IGFET drawn similarly to IGFETs (400A and/or 400B) illustrate IGFETs that have a plurality of horizontally disposed and vertically aligned channel regions.

Referring now to FIG. 5, an SRAM memory cell is set forth in a circuit schematic diagram and given the general reference character 500.

SRAM memory cell may include IGFETs (P510, P520, N510, N520, N530, and N540). IGFETs (P510, P520, N510, N520, N530, and N540) may each have a plurality of horizontally disposed and vertically aligned channel regions that are surrounded by a control gate.

IGFET P510 may have a source terminal electrically connected to receive a memory cell power supply potential Vcell, a drain terminal electrically connected to a storage node N1, and a gate terminal electrically connected to storage node N2. IGFET P520 may have a source terminal electrically connected to receive a memory cell power supply potential Vcell, a drain terminal electrically connected to a storage node N2, and a gate terminal electrically connected to storage node N1. IGFETs (P510 and P520) may be P-channel IGFETs, also known as P-type IGFETs.

IGFET N510 may have a source terminal electrically connected to a reference potential VSS, a drain terminal electrically connected to storage node N1, and a gate terminal electrically connected to storage node N2. IGFET N520 may have a source terminal electrically connected to a reference potential VSS, a drain terminal electrically connected to storage node N2, and a gate terminal electrically connected to storage node N1. IGFET N530 may have a first source/drain terminal electrically connected to bit line BL, a second source/drain terminal electrically connected to storage node N1 and a gate terminal electrically connected to a word line WL. In this way, IGFET N530 can form a controllable impedance path between bit line BL and storage node N1. IGFET N540 may have a first source/drain terminal electrically connected to complementary bit line BLb, a second source/drain terminal electrically connected to storage node N2 and a gate terminal electrically connected to a word line WL. In this way, IGFET N540 can form a controllable impedance path between complementary bit line BLb and storage node N2. IGFETs (N510, N520, N530 and N540) may be N-channel IGFETs, also known as N-type IGFETs.

Referring now to FIG. 6, an integrated circuit device portion including an SRAM portion having end power supply circuits providing a write assist according to an embodiment is set forth in a circuit schematic diagram and given the general reference character 600.

Integrated circuit device portion 600 may correspond to a portion of integrated circuit device of FIG. 2. In particular, integrated circuit device 600 may have similar constituents as integrated circuit device 200, such constituents may begin with the general reference character "6" instead of "2".

Integrated circuit device portion 600 illustrates a more detailed embodiment of SRAM array 290 and end power supply select circuits (260 and 270) of integrated circuit device 200 of FIG. 2.

Integrated circuit device portion 600 can include a row decoder circuit 650, end power select circuits (660 and 670), and an SRAM array 690.

Row decoder circuit **650** may receive a word line enable signal WLEN, row address signals (RA1 to RAn), where n is the number of row addresses, and power supply potential VDD and may provide word line signals to word lines **652** (WL1 to WLx), where x is the number of word lines in SRAM array **690**.

End power supply select circuit **660** may receive power supply potential VDD and end power supply control signals (EP11 and EP12) and may provide a power supply potential to SRAM array power lines **662**. SRAM array power lines **662** may provide power to SRAM cells **692** in SRAM array **690**.

End power supply select circuit **670** may receive power supply potential VDD and end power supply control signals (EP21 and EP22) and may provide a power supply potential to SRAM array power lines **662**. SRAM array power lines **662** may provide power to SRAM cells **692** in SRAM array **690**.

SRAM array **690** may be electrically connected to word lines WL1 to WLx and SRAM array power lines **662**.

SRAM array **690** may include SRAM memory cells **692**. Each SRAM memory cell **692** may be located at a cross point of an individual word line (WL1 to WLx) and a bit line pair (not shown in order to avoid unduly cluttering the figure). Bit line pairs may be disposed essentially orthogonal to word lines (WL1 to WLx). Each memory cell **692** may be electrically connected to an array power line **662**. Array power lines **662** may be arranged substantially in parallel to bit line pairs and substantially perpendicular to word lines (WL1 to WLx).

End power select circuit **660** may include selection circuits **664**. Each power enable circuit **664** are electrically connected to receive power supply potential VDD and end power supply control signals (EP11 and EP12). Each selection circuit **664** is electrically connected to a plurality of array power lines **662**. Each selection circuit **664** can include IGFETs (P662 and P664). IGFET P662 can receive end power supply control signal EP11 at a control gate terminal. IGFET P662 may have a source terminal electrically connected to receive power supply potential VDD and a drain terminal electrically connected to a plurality of array power supply lines **662**. IGFET P664 can receive end power supply control signal EP12 at a control gate terminal. IGFET P664 may have a source terminal electrically connected to receive power supply potential VDD and a drain terminal electrically connected to a plurality of array power supply lines **662**. IGFETs (P662 and P664) may be P-channel IGFETs. End power select circuit **660** may provide a controllable impedance path between power supply potential VDD and a group of power supply lines **662** at a first end of SRAM array **690**.

End power select circuit **670** may include selection circuits **674**. Each power enable circuit **674** are electrically connected to receive power supply potential VDD and end power supply control signals (EP21 and EP22). Each selection circuit **674** is electrically connected to a plurality of array power lines **662**. Each selection circuit **674** can include IGFETs (P672 and P674). IGFET P672 can receive end power supply control signal EP21 at a control gate terminal. IGFET P672 may have a source terminal electrically connected to receive power supply potential VDD and a drain terminal electrically connected to a plurality of array power supply lines **672**. IGFET P674 can receive end power supply control signal EP22 at a control gate terminal. IGFET P674 may have a source terminal electrically connected to receive power supply potential VDD and a drain terminal electrically connected to a plurality of array power supply lines

672. IGFETs (P672 and P674) may be P-channel IGFETs. End power select circuit **670** may provide a controllable impedance path between power supply potential VDD and a group of power supply lines **662** at a second end of SRAM array **690**.

It is understood that power supply lines **662** and word lines **652** may be essentially perpendicular to each other, however, word lines **652** do not contact power supply lines **662** and are formed by a different layer of interconnect material formed on a different plane.

Referring now to FIG. 7, an array power control circuit is set forth in a circuit schematic diagram and given the general reference character **700**. Array power control circuit **700** may be used as array power control circuit **240** on integrated circuit device **200** of FIG. 2.

Array power control circuit **700** may receive assist enable signal AEN, row address (RAn and RA(n-1)) and write signal WR and may provide end power supply control signals (EP11, EP21, EP12, and EP22). Row address RAn may be the most significant row address bit received by row decoder circuit (**250** and **650**) of FIGS. 2 and 6, respectively. Row address RA(n-1) may be the next most significant row address bit received by row decoder circuit (**250** and **650**) of FIGS. 2 and 6, respectively.

Array power control circuit **700** may include logic gates (G710, G716, G720, G726, G730, G736, G740, G746, G750, G756, G760, and G766).

Logic gate G710 may receive row address signal RAn at an input terminal and may provide an output at an output terminal. Logic gate G710 may be an inverter logic gate. Logic gate G716 may receive row address signal RA(n-1) at an input terminal and may provide an output at an output terminal. Logic gate G716 may be an inverter logic gate. Logic gate G720 may receive the output of logic gate G710 and the output of logic gate G716 at respective input terminals and may provide an output. Logic gate G720 may be a NAND logic gate. Logic gate G726 may receive row address signal RAn and row address signal RA(n-1) at respective input terminals and may provide an output. Logic gate G726 may be a NAND logic gate.

Logic gate G730 may receive assist enable signal AEN, write signal WR, and the output of logic gate G710 at respective input terminals and may provide an output. Logic gate G730 may be a NAND logic gate. Logic gate G736 may receive the output of logic gate G730 at an input and may provide end power supply control signal EP11 at an output terminal. Logic gate G736 may be an inverter logic gate.

Logic gate G740 may receive assist enable signal AEN, write signal WR, and row address signal RAn at respective input terminals and may provide an output. Logic gate G740 may be a NAND logic gate. Logic gate G746 may receive the output of logic gate G740 at an input and may provide end power supply control signal EP21 at an output terminal. Logic gate G746 may be an inverter logic gate.

Logic gate G750 may receive assist enable signal AEN, write signal WR, and the output of logic gate G720 at respective input terminals and may provide an output. Logic gate G750 may be a NAND logic gate. Logic gate G756 may receive the output of logic gate G750 at an input and may provide end power supply control signal EP22 at an output terminal. Logic gate G756 may be an inverter logic gate.

Logic gate G760 may receive assist enable signal AEN, write signal WR, and the output of logic gate G726 at respective input terminals and may provide an output. Logic gate G760 may be a NAND logic gate. Logic gate G766 may receive the output of logic gate G760 at an input and may

provide end power supply control signal EP12 at an output terminal. Logic gate G766 may be an inverter logic gate.

The operation of integrated circuit device will now be discussed with reference to FIGS. 1 through 9.

FIG. 8 is a timing logic diagram of the integrated circuit device when the power supply potential is below an assist potential Vassist.

Referring to FIG. 8, power supply potential VDD has a potential value that is less than an assist potential Vassist. When power supply potential VDD is less than assist potential Vassist, power supply level detection circuit 235 (FIG. 2) may provide an assist enable signal AEN having a logic high level (assist enable logic level). In this way, write assist may be enabled to increase the write margin to SRAM array 290 at lower operating levels (i.e. power supply potentials).

At time T1, write signal WR (i.e. write enable signal) may go high, indicating a write operation to SRAM array 290. With write signal WR at a logic high and assist enable signal AEN at a logic high, logic gates (G730, G740, G750, and G760) in array power control circuit 700 of FIG. 7 may be enabled. In this way, power control circuit 700 may select IGFETs (P662, P664, P672 and/or P674) in a selection circuit (664 or 674) in respective end power select circuit (660 or 670) (FIG. 6) in accordance with a location of a selected word line (WL1 to WLx) in array 690.

Also at time T1, with write signal WR at a logic high level, indicating an initiation of a write operation in which data may be written to a selected cell or selected cells in SRAM array (690, 290). With row address signal RAn at a logic low level, and row address signal RA(n-1) at a logic low level, the SRAM cell that is being written to may be in the top quadrant as illustrated in SRAM array 690 of FIG. 6. In this case, power control circuit 700 may provide end power control signals (EP21 and EP22) having a logic low level and end power control signals (EP11 and EP12) having a logic high level. In this way, selection circuits in end power select circuit 670 may be enabled by placing IGFETs (P672 and P674) in low impedance states and end power select circuit 660 may be disabled by placing IGFETs (P662 and P664) in high impedance states. By doing so, a resistance between power supply potential VDD and a selected memory cell 692 may be increased by providing the power supply potential through a longer portion of the power supply line 662. By doing so, the supply potential at the selected memory cell may be at a lower potential due to an I-R (current times resistance) drop and data may be written to a selected memory cell 692 more reliably, as the pull up capabilities of the p-type IGFETs (P510 and P520) may not be able to overcome the pull down capabilities of N-channel IGFETs (N530, N510) (for example when writing a zero in the memory cell (500, 692)) or N-channel IGFETs (N540, N520) (for example when writing a one in the memory cell (500, 692)). It should be noted that the word line potential at the select gate IGFETs (N530 and N540) may still be substantially at a full power supply potential VDD as the row decoder circuit 250 may be powered locally and in proximity to the row decoder circuit 250 and the select gate IGFETs (N530 and N540) may essentially have a boosted control gate potential as compared to the power potential driving the selected memory cell 692.

At time T2, row address RAn may be low and row address RA(n-1) may transition to a logic high level. With row address signal RAn at a logic high level, and row address signal RA(n-1) at a logic high level, a word line (WL1 to WLx) in the top middle quadrant may become active and the SRAM cell that is being written to may be in the top middle

quadrant as illustrated in SRAM array 690 of FIG. 6. In this case, power control circuit 700 may provide end power control signal EP21 having a logic low level and end power control signals (EP22, EP11, and EP12) having a logic high level. In this way, selection circuits in end power select circuit 670 may be enabled by placing IGFET P672 in low impedance states and may place IGFET P674 in a high impedance state. End power select circuit 660 may be disabled by placing IGFETs (P662 and P664) in high impedance states. By doing so, a resistance between power supply potential VDD and a selected memory cell 692 may be increased by providing the power supply potential through a longer portion of the power supply line 662. It is noted that the distance between the between power supply potential VDD and a selected memory cell 692 may be substantially less than the case at time T1 as the selected memory cell in the upper middle quadrant is closer to the enabled end power select circuit 670. This can substantially reduce the resistance (impedance) of the power supply line 662. However, by turning off IGFET P674, the impedance (resistance) of the end power select circuit 670 may be substantially increased to provide an I-R drop more consistent to the I-R drop provided at time T1 to the selected memory cell 692. By doing so, the supply potential at the selected memory cell may be at a lower potential due to an I-R (current times resistance) drop and data may be written to a selected memory cell 692 more reliably, as the pull up capabilities of the p-type IGFETs (P510 and P520) may not be able to overcome the pull down capabilities of N-channel IGFETs (N530, N510) (for example when writing a zero in the memory cell (500, 692)) or N-channel IGFETs (N540, N520) (for example when writing a one in the memory cell (500, 692)). It should be noted that the word line potential at the select gate IGFETs (N530 and N540) may still be substantially at a full power supply potential VDD as the row decoder circuit 250 may be powered locally and in proximity to the row decoder circuit 250 and the select gate IGFETs (N530 and N540) may essentially have a boosted control gate potential as compared to the power potential driving the selected memory cell 692.

At time T3, row address RA(n-1) may be high and row address RA(n) may transition to a logic high level. With row address signal RAn at a logic high level, and row address signal RA(n-1) at a logic high level, the SRAM cell that is being written to may be in the bottom quadrant as illustrated in SRAM array 690 of FIG. 6. In this case, power control circuit 700 may provide end power control signals (EP21 and EP22) having a logic high level and end power control signals (EP11 and EP12) having a logic low level. In this way, selection circuits in end power select circuit 660 may be enabled by placing IGFETs (P662 and P664) in low impedance states and end power select circuit 670 may be disabled by placing IGFETs (P672 and P674) in high impedance states. By doing so, a resistance between power supply potential VDD and a selected memory cell 692 may be increased by providing the power supply potential through a longer portion of the power supply line 662. By doing so, the supply potential at the selected memory cell may be at a lower potential due to an I-R (current times resistance) drop and data may be written to a selected memory cell 692 more reliably, as the pull up capabilities of the p-type IGFETs (P510 and P520) may not be able to overcome the pull down capabilities of N-channel IGFETs (N530, N510) (for example when writing a zero in the memory cell (500, 692)) or N-channel IGFETs (N540, N520) (for example when writing a one in the memory cell (500, 692)). It should be noted that the word line potential at the select gate IGFETs

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(N530 and N540) may still be substantially at a full power supply potential VDD as the row decoder circuit 250 may be powered locally and in proximity to the row decoder circuit 250 and the select gate IGFETs (N530 and N540) may essentially have a boosted control gate potential as compared to the power potential driving the selected memory cell 692.

At time T4, row address RAn may be at a logic high and row address RA(n-1) may transition to a logic low level. With row address signal RAn at a logic high level, and row address signal RA(n-1) at a logic low level, a word line (WL1 to WLx) in the bottom middle quadrant may become active and the SRAM cell that is being written to may be in the bottom middle quadrant as illustrated in SRAM array 690 of FIG. 6. In this case, power control circuit 700 may provide end power control signal EP11 having a logic low level and end power control signals (EP22, EP21, and EP12) having a logic high level. In this way, selection circuits in end power select circuit 660 may be enabled by placing IGFET P662 in low impedance states and may place IGFET P664 in a high impedance state. End power select circuit 670 may be disabled by placing IGFETs (P672 and P674) in high impedance states. By doing so, a resistance between power supply potential VDD and a selected memory cell 692 may be increased by providing the power supply potential through a longer portion of the power supply line 662. It is noted that the distance between the power supply potential VDD and a selected memory cell 692 may be substantially less than the case at time T3 as the selected memory cell in the lower middle quadrant is closer to the enabled end power select circuit 660. This can substantially reduce the resistance (impedance) of the power supply line 662. However, by turning off IGFET P664, the impedance (resistance) of the end power select circuit 660 may be substantially increased to provide an I-R drop more consistent to the I-R drop provided at time T3 to the selected memory cell 692. By doing so, the supply potential at the selected memory cell may be at a lower potential due to an I-R (current times resistance) drop and data may be written to a selected memory cell 692 more reliably, as the pull up capabilities of the p-type IGFETs (P510 and P520) may not be able to overcome the pull down capabilities of N-channel IGFETs (N530, N510) (for example when writing a zero in the memory cell (500, 692)) or N-channel IGFETs (N540, N520) (for example when writing a one in the memory cell (500, 692)). It should be noted that the word line potential at the select gate IGFETs (N530 and N540) may still be substantially at a full power supply potential VDD as the row decoder circuit 250 may be powered locally and in proximity to the row decoder circuit 250 and the select gate IGFETs (N530 and N540) may essentially have a boosted control gate potential as compared to the power potential driving the selected memory cell 692.

By providing consistent power supply VDD I-R drops regardless as to which quadrant (region), the selected memory cell is located a consistent memory cell power supply potential Vcell may be provided to a selected memory cell 692 regardless as to location in SRAM memory array 290, write margins may be more optimal and consistent to provide improved overall operating margins. In order to provide a consistent resistance (impedance) between a selected memory cell 692 and power supply VDD, the on impedance of IGFETs (P662 and P672) may be substantially less than the on impedance of IGFETs (P664 and P674). This may be done by providing IGFETs (662 and 672) with a shorter control gate length L and/or a wider control gate width W as compared to the control gate length and/or

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control gate width of IGFETs (664 and 674). By doing so, the resistance/impedance between a selected memory cell 692 and power supply VDD may be incrementally modified without having detrimental differences in memory cell power supply potential Vcell due to location of the selected memory cell 692 in the SRAM memory array (290 and 690).

It is noted that in the above example, the memory array 690 may be divided into quadrants or regions as determined by the two most significant row address signals (RAn and RA(n-1)). A substantially consistent memory cell power supply potential Vcell may be provided during a write operation to a selected memory cell 692 regardless as to the quadrant or region in which the selected memory cell 692 may be located. In the above example a first region/quadrant may be the area illustrated of memory array 690 when row address RAn=0 and row address RA(n-1)=0, a second region/quadrant may be the area illustrated of memory array 690 when row address RAn=0 and row address RA(n-1)=1, a third region/quadrant may be the area illustrated of memory array 690 when row address RAn=1 and row address RA(n-1)=0, and a fourth region/quadrant may be the area illustrated of memory array 690 when row address RAn=1 and row address RA(n-1)=1.

FIG. 9 is a logic timing diagram of the integrated circuit device when the power supply potential VDD is greater than assist potential Vassist.

Referring now to FIG. 9, power supply potential VDD is above assist potential Vassist. When power supply potential VDD is greater than assist potential Vassist, power supply level detection circuit 235 (FIG. 2) may provide an assist enable signal AEN having a logic low level (assist disable logic level). In this way, write assist may be disabled as the write assist may not be needed to provide increased write margin due to the increased current driving capabilities of of N-channel IGFETs (N530, N510) in SRAM memory cell 500.

With assist enable signal AEN at a logic low level, logic gates (G730, G740, G750, and G760) in array power control circuit 700 of FIG. 7 may be disabled and end power control signals (EP11, EP21, EP12, and EP22) may all be forced to a logic low level. With end power control signals (EP11, EP21, EP12, and EP22) at a logic low level, IGFETs (P662, P664, P672, and P674) in selection circuits (664 and 674) in respective end power select circuits (660 and 670) may be in a low impedance state and IR drops to memory power supply potential Vcell at a selected memory cell 692 may be reduced.

It should be noted, during a read operation in which data is provided from a selected memory cell 692, write enable signal WR is at a logic low level. In this way, power control circuit 700 provides end power control signals (EP11, EP21, EP12, and EP22) may all be forced to a logic low level. With end power control signals (EP11, EP21, EP12, and EP22) at a logic low level, IGFETs (P662, P664, P672, and P674) in selection circuits (664 and 674) in respective end power select circuits (660 and 670) may be in a low impedance state and IR drops to memory power supply potential Vcell at a selected memory cell 692 may be reduced and the speed of a read operation may not be inhibited.

Referring now to FIG. 10, an integrated circuit device portion including an SRAM portion having end power supply circuits providing a write assist according to an embodiment is set forth in a circuit schematic diagram and given the general reference character 1000.

Integrated circuit device portion 1000 may include constituents that are the same as constituents of integrated circuit portion 600, such constituents are given the same

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reference character. In particular, integrated circuit device **1000** may differ from integrated circuit device portion **600** in that integrated circuit device portion **1000** may have end power select circuits (**1060** and **1070**) that include select circuits (**1064** and **1074**) having a single IGFET (**P1062** and **P1072**) respectively. IGFET **P1062** may receive end power control signal **EP11** at a gate terminal and IGFET **P1072** may receive end power control signal **EP21** at a gate terminal.

Referring now to FIG. **11**, an array power control circuit is set forth in a circuit schematic diagram and given the general reference character **1100**. Array power control circuit **1100** may be used as array power control circuit **240** on integrated circuit device **200** of FIG. **2** in conjunction with integrated circuit device portion **1000**. Array power control circuit **1100** may include the same constituents as array power control circuit **700** for generating end power control signals (**EP11** and **EP21**) as set forth above.

FIG. **12** is a logic timing diagram of the integrated circuit device implementing integrated circuit device portion **1000** when the power supply potential is below an assist potential Vassist.

Referring to FIG. **12**, power supply potential **VDD** has a potential value that is less than an assist potential Vassist. When power supply potential **VDD** is less than assist potential Vassist, power supply level detection circuit **235** (FIG. **2**) may provide an assist enable signal **AEN** having a logic high level (assist enable logic level). In this way, write assist may be enabled to increase the write margin to SRAM array **290** at lower operating levels.

As shown in FIG. **12**, end power select circuit **1060** may be enabled (IGFET **1062** is in a low impedance state) and end power select circuit **1070** may be disabled (IGFET **1072** is in a high impedance state) when the selected memory cell **692** is in the lower half of the SRAM array **690**. End power select circuit **1070** may be enabled (IGFET **1072** is in a low impedance state) and end power select circuit **1060** may be disabled (IGFET **1062** is in a high impedance state) when the selected memory cell **692** is in the upper half of the SRAM array **690**. By doing so, a resistance between power supply potential **VDD** and a selected memory cell **692** may be increased by providing the power supply potential through a longer portion of the power supply line **662**. In this way, the supply potential at the selected memory cell may be at a lower potential due to an I-R (current times resistance) drop and data may be written to a selected memory cell **692** more reliably, as the pull up capabilities of the p-type IGFETs (**P510** and **P520**) may not be able to overcome the pull down capabilities of N-channel IGFETs (**N530**, **N510**) (for example when writing a zero in the memory cell (**500**, **692**)) or N-channel IGFETs (**N540**, **N520**) (for example when writing a one in the memory cell (**500**, **692**)). It should be noted that the word line potential at the select gate IGFETs (**N530** and **N540**) may still be substantially at a full power supply potential **VDD** as the row decoder circuit **250** may be powered locally and in proximity to the row decoder circuit **250** and the select gate IGFETs (**N530** and **N540**) may essentially have a boosted control gate potential as compared to the power potential driving the selected memory cell **692**.

FIG. **13** is a timing logic diagram of the integrated circuit device when the power supply potential **VDD** is greater than assist potential Vassist.

Referring now to FIG. **13**, power supply potential **VDD** is above assist potential Vassist. When power supply potential **VDD** is greater than assist potential Vassist, power supply level detection circuit **235** (FIG. **2**) may provide an assist

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enable signal **AEN** having a logic low level (assist disable logic level). In this way, write assist may be disabled as the write assist may not be needed to provide increased write margin due to the increased current driving capabilities of N-channel IGFETs (**N530**, **N510**) in SRAM memory cell **500**.

With assist enable signal **AEN** at a logic low level, logic gates (**G730** and **G740**) in array power control circuit **1100** of FIG. **11** may be disabled and end power control signals (**EP11** and **EP21**) may be forced to a logic low level. With end power control signals (**EP11** and **EP21**) at a logic low level, IGFETs (**P1062** and **P1072**) in selection circuits (**1064** and **1074**) in respective end power select circuits (**1060** and **1070**) may be in a low impedance state and IR drops to memory power supply potential Vcell at a selected memory cell **692** may be reduced.

Referring now to FIG. **14**, a circuit schematic diagram of power supply level detection circuit **235** according to an embodiment is set forth.

Power supply detection circuit **235** includes a logic gate **1410**, IGFET **P1402**, resistors (**R1402** and **R1404**), a capacitor **C1402**, and an amplifier circuit **A1402**.

Logic gate **1410** receives comparison enable signal **EN** at an input terminal and provides an output. Logic gate **1410** may be an inverter logic gate. IGFET **P1402** may receive power supply potential **VDD** at a source terminal, and may receive the output of logic gate **1410** at a control gate terminal, and may have a drain terminal connected to a first terminal of resistor **R1402**. IGFET **P1402** may be a p-type IGFET. Resistor **R1402** may have a second terminal connected to a node **N1400**. Resistor **R1404** may have a first terminal connected to reference potential **VSS** (ground reference potential) and a second terminal connected to node **N1400**. Capacitor **C1402** may have a first terminal connected to reference potential **VSS** and a second terminal connected to node **N1400**. Amplifier circuit **A1402** may be a differential amplifier circuit and may have a negative input terminal connected to node **N1400** and a positive input terminal connected to receive a reference potential Vref, and may provide assist enable signal **AEN** at an output terminal.

The operation of power supply detection circuit **235** will now be explained. When comparison enable signal **EN** is at a logic low level (comparison disable logic level), logic gate **1410** may provide a logic high output. With a logic high level at the control gate of IGFET **P1402**, IGFET **P1402** may provide a high impedance path between power supply potential **VDD** and the first terminal of resistor **R1402**. In this way, resistor **R1402** may pull node **N1400** to reference potential **VSS**. Because reference potential **VSS** has a lower potential than reference potential Vref, amplifier circuit **A1402** provides an assist enable signal having a logic high level. It is noted that when comparison enable signal **EN** is at a logic low, current is prevented from flowing from power supply potential **VDD** to reference potential **VSS** through resistors (**R1402** and **R1404**), thus power usage may be reduced.

When comparison enable signal **EN** is at a logic high level (comparison enable logic level), logic gate **1402** may provide a logic low output. With a logic low level at the control gate of IGFET **P1402**, IGFET **P1402** may provide a low impedance path between power supply potential and the first terminal of resistor **R1402**. In this way, resistors (**R1402** and **R1404**) may provide a voltage divider circuit to provide a potential at node **N1400** that is proportional to power supply potential **VDD**. Reference potential Vref has been selected to provide a potential that is higher than the potential at node **N1400** when write assist is needed and lower than the

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potential at node **N1400** when write assist is not needed. Thus, when write assist is needed, amplifier circuit **A1402** provides assist enable signal **AEN** having a logic high level (assist enable logic level) and when write assist is not needed, amplifier circuit **A1402** provides assist enable signal **AEN** having a logic low level (assist disable logic level).

Referring now to FIG. 15, a flow chart diagram illustrating a method of operating an integrated circuit device according to an embodiment is set forth and given the general reference character **1500**.

Method **1500** includes steps (**S1502**, **S1506**, **S1510**, **S1512**, **S1516**, **S1517**, **S1518**, **S1520**, **S1524**, **S1528**, **S1532**, **S1536**, and **S1540**).

Step **1502**, a determination is made as to whether power supply detection circuit **235** is enabled.

At step **1506**, a determination is made as to whether power supply potential **VDD** is less than an assist potential **Vassist**.

At step **1510**, a determination is made as whether integrated circuit device **100** is executing a write operation.

At step **1512**, all IGFETs (**P662**, **P664**, **P672**, and **P674**) in selection circuits (**664** and **674**) in respective end power select circuits (**660** and **670**) are placed in a low impedance state to provide minimal resistance between power supply potential **VDD** and memory cells **692**.

At step **S1516**, a value of the most significant row address signal **RA_n** is determined. By doing so, one of end power select circuits (**660** and **670**) may be enabled while the other of end power select circuit (**660** and **670**) may have respective IGFETs ((**P662** and **P664**) or (**P672**, and **P674**)) placed in a high impedance state. For example, if row address signal **RA_n** is "0", end power select circuit **660** has IGFETs (**P662** and **P664**) placed in a high impedance state and if row address signal **RA_n** is "1", end power select circuit **670** has IGFETs (**P672** and **P674**) placed in a high impedance state.

At step **S1517**, IGFETs (**P662** and **P664**) in selection circuit **664** of end power select circuit **660** are turned off and therefore placed in a high impedance state.

At step **S1518**, IGFETs (**P672** and **P674**) in selection circuit **674** of end power select circuit **670** are turned off and therefore placed in a high impedance state.

At step **S1520** and at step **S1524**, a value of next most significant row address **RA_(n-1)** is determined.

At step **S1528**, IGFET **P674** is turned off and placed in a high impedance state, and IGFET **P672** is placed in a low impedance state (turned on). At this step, it has been determined that the selected memory cell is closer to the middle than the end of the array.

At step **S1532**, IGFETs (**P672** and **P674**) are turned on a placed in a low impedance state. At this step, it has been determined that the selected memory cell is closer to the end of the array than the middle of the array.

At step **S1536**, IGFET **P664** is turned off and placed in a high impedance state (turned off), and IGFET **P662** is placed in a low impedance state (turned on). At this step, it has been determined that the selected memory cell is closer to the middle than the end of the array.

At step **S1540**, IGFETs (**P662** and **P664**) are turned on a placed in a low impedance state (turned on). At this step, it has been determined that the selected memory cell is closer to the end of the array than the middle of the array.

During a read operation, write enable signal **WR** is at a logic low level.

Although row address **RA_n** has been referred to as a most significant row address, it may refer to any row address that is used as a most significant row address of an array of memory cell and may not be a most significant row address

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on a data sheet or in a memory that has many memory banks or sections in which banks or sections may be selected by a more significant address.

Other electrical apparatus other than semiconductor devices may benefit from the invention.

While various particular embodiments set forth herein have been described in detail, the present invention could be subject to various changes, substitutions, and alterations without departing from the spirit and scope of the invention.

Accordingly, the present invention is intended to be limited only as defined by the appended claims.

What is claimed is:

1. A method of operating an integrated circuit device including the steps of:

enabling a power control circuit to provide power control signals according to at least one row address signal when an assist enable signal has an assist enable logic level and the integrated circuit device is in a write mode of operation wherein data is written to at least one selected SRAM memory cell; and

selectively controlling at least one impedance path between a power supply potential and an array power line electrically connected to provide power to the at least one selected SRAM memory cell in a memory array in response to the at least one power control signal,

wherein each at least one impedance path between the power supply potential and the array power line includes a plurality of channels between a source terminal and a drain terminal of an insulated gate field effect transistor (IGFET), the plurality of channels disposed in a first direction and substantially aligned in a second direction substantially perpendicular to the first direction and having a control gate structure that contiguously surrounds the plurality of channels.

2. The method of operating an integrated circuit device of claim 1, further including the step of:

providing the assist enable signal having the assist enable logic level in response to a power supply potential being below an assist potential.

3. The method of operating an integrated circuit device of claim 2, further including the step of:

providing the assist enable signal having an assist disable logic level in response to the power supply potential being above the assist potential.

4. The method of operating an integrated circuit device of claim 1, further including:

providing a low impedance path between the power supply potential and the array power line on a first end of the memory array and providing a high impedance path between the power supply potential and the array power line on a second end of the memory array opposite to the first end of the memory array when the at least one selected SRAM memory cell is closer to the second end of the memory array.

5. The method of operating an integrated circuit device of claim 4, further including the step of:

providing a low impedance path between the power supply potential and the array power line on the second end of the memory array and providing a high impedance path between the power supply potential and the array power line on the first end of the memory array opposite to the second end of the memory array when the at least one selected SRAM memory cell is closer to the first end of the memory array.

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6. The method of operating an integrated circuit device of claim 4, wherein:

the memory array is arranged in a plurality of rows and columns of memory cells and the array power line is electrically connected to a column of memory cells to provide a memory cell power supply to each of the memory cells in the column of memory cells.

7. The method of operating an integrated circuit device of claim 6, wherein:

the IGFET is a p-channel IGFET.

8. The method of operating the integrated circuit device of claim 1, further including:

enabling a power control circuit to provide power control signals having the same logic level in a read mode of operation wherein data is read from at least one selected memory cell.

9. The method of operating the integrated circuit device of claim 1, wherein:

the integrated device includes a processor circuit portion and a SRAM circuit portion, the processor portion and the SRAM circuit portion are commonly powered by the power supply potential, the SRAM circuit portion includes the at least one selected SRAM memory cell.

10. The method of operating the integrated circuit device of claim 9, wherein:

the processor portion and the SRAM circuit portion are connected by a bus for carrying data between the processor portion and the SRAM circuit portion.

11. A method of operating an integrated circuit device including the steps of:

enabling a power control circuit to provide power control signals according to at least one row address signal when an assist enable signal has an assist enable logic level and the integrated circuit device is in a write mode of operation, wherein data is written to at least one selected SRAM memory cell;

selectively controlling at least one impedance path between a power supply potential and an array power line electrically connected to provide power to the at least one selected SRAM memory cell in a memory array in response to the at least one power control signal;

providing a low impedance path between the power supply potential and the array power line on a first end of the memory array and providing a high impedance path between the power supply potential and the array power line on a second end of the memory array opposite to the first end of the memory array when the at least one selected SRAM memory cell is closer to the second end of the memory array,

providing a low impedance path between the power supply potential and the array power line on the second end of the memory array and providing a high impedance path between the power supply potential and the array power line on the first end of the memory array opposite to the second end of the memory array when the at least one selected SRAM memory cell is closer to the first end of the memory array,

providing the low impedance path between the power supply potential and the array power line on the first end of the memory array and the second end of the memory array when the assist enable signal has an assist disable logic level,

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wherein each at least one impedance path between the power supply potential and the array power line includes a plurality of channels between a source terminal and a drain terminal of an insulated gate field effect transistor (IGFET), the plurality of channels disposed in a first direction and substantially aligned in a second direction substantially perpendicular to the first direction and having a control gate structure that contiguously surrounds the plurality of channels.

12. The method of operating the integrated circuit device of claim 11, further including the step of:

providing the assist enable signal having the assist enable logic level in response to a power supply potential being below an assist potential.

13. The method of operating the integrated circuit device of claim 12, further including the step of:

providing the assist enable signal having an assist disable logic level in response to the power supply potential being above the assist potential.

14. The method of operating an integrated circuit device of claim 12, further including the step of:

comparing the power supply potential to a reference potential to provide the assist enable signal.

15. The method of operating an integrated circuit device of claim 14, further including the step of:

providing a comparison enable signal having a comparison enable logic level to enable the step of comparing the power supply potential to the reference potential.

16. The method of operating an integrated circuit device of claim 14, wherein:

providing a comparison enable signal having a comparison disable logic level to disable the step of comparing the power supply potential to a reference potential.

17. The method of operating an integrated circuit device of claim 16, further including the step of:

providing the assist enable signal having the assist disable logic level in response to the comparison enable signal having the comparison disable logic level.

18. The method of operating the integrated circuit device of claim 14, wherein:

the comparing the power supply potential to the reference potential to provide the assist enable signal generates the assist enable signal having the assist enable logic level in response to the power supply potential being less than an assist potential and generates the assist enable signal having the assist disable logic level in response to the power supply potential being greater than the assist potential.

19. The method of operating the integrated circuit device of claim 11, wherein:

the integrated device includes a processor circuit portion and a SRAM circuit portion, the processor portion and the SRAM circuit portion are commonly powered by the power supply potential, the SRAM circuit portion includes the at least one selected SRAM memory cell.

20. The method of operating the integrated circuit device of claim 19, wherein:

the processor portion and the SRAM circuit portion are connected by a bus for carrying data between the processor portion and the SRAM circuit portion.

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